

Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

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Course: *Machine Learning in Econometrics*

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Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the paper.

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Replication files are available in the [GitHub repository](#) for this project.

1 Introduction

2 Empirical Strategy and Identification

- Target parameters: ATE, ATT, GATT (possible differences in causal forest and aipw lasso estimation) (show math and explain shortly)

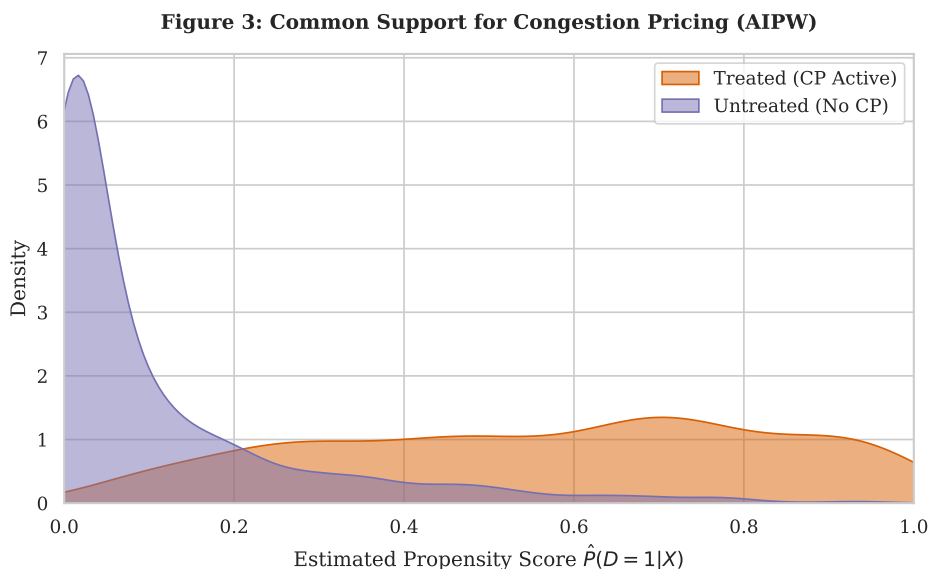


Figure 1: Common Support for Congestion Pricing (AIPW)

3 Proposed Estimator

- Use AIPW and Causal Forest instead of IPW or Reg. adj.
- Explain how they work and differences

4 Theoretical Properties

- Proof of Neyman Orthogonality (is it the same for both?)
- Not yet sure about other properties

5 Data Preparation

- no missing values or absurd numbers - check for skewed distributions (plot in appendix)
- log transform variables where it makes economic sense and where it solves skewness -
- Time invariant variables and creation of Mundlak device in order to account for panel structure (national climate pact removed because not changing)

6 K-means clustering

- determination of two clusters necessary to make heterogeneity analysis with many co-variates viable - optimal $k = 2$ (interpretability and silhouette analysis) - Cluster 1: urban metropolises, Cluster 2: sprawling regional hubs - describe them based on the cluster differences plot

Table 1: Distribution of Climate Policy Treatments by Urban Typology

Urban Typology	Never Treated	CP Only	LEZ Only	Synergy (CP $\times$ LEZ)	Total
Dense Metropolises (1)	12	27	16	24	79
Regional Hubs (0)	61	13	26	4	104
Total	73	40	42	28	183

Notes: Table displays the count of unique cities. A city is classified as “Ever Treated” if Congestion Pricing was active in any year during the sample period.

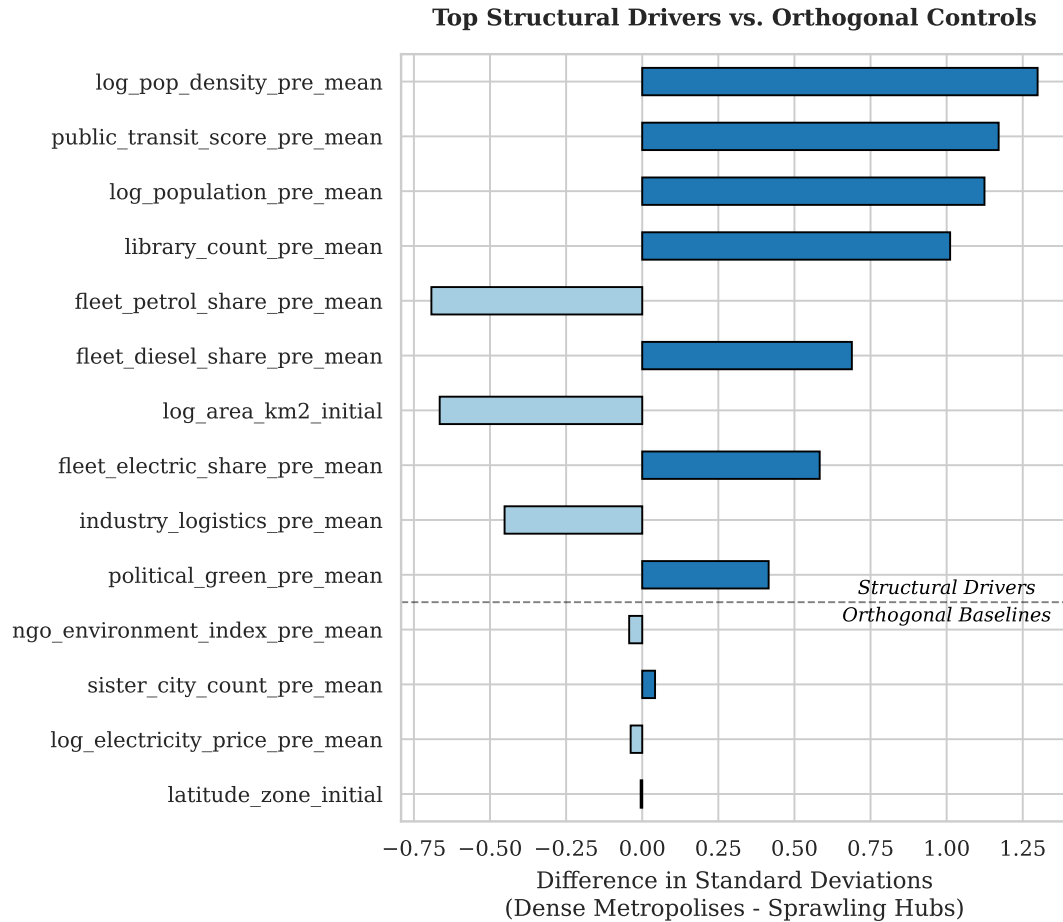


Figure 2: Differences between the two clusters.

Table 2: Double Machine Learning Estimates of Urban Climate Policies (Categorical Regime)

Policy	(1) OLS Baseline	(2) AIPW (Lasso)	(3) Causal Forest
Panel A: Global Average Treatment Effect (ATE)			
Congestion Pricing (CP)	–	–0.2991*** ($p = 0.002$)	–0.3829*** ($p = 0.000$)
Low Emission Zone (LEZ)	–	–0.1001 ($p = 0.157$)	–0.1528* ($p = 0.061$)
Synergy (CP $\times$ LEZ)	–	–1.0725*** ($p = 0.000$)	–0.8590*** ($p = 0.000$)
Panel B: Average Treatment Effect on the Treated (ATT)			
Congestion Pricing (CP)	–	–0.3190** ($p = 0.013$)	–0.3904*** ($p = 0.000$)
Low Emission Zone (LEZ)	–	–0.1085 ($p = 0.123$)	–0.1627* ($p = 0.056$)
Synergy (CP $\times$ LEZ)	–	–1.3667*** ($p = 0.000$)	–1.2622*** ($p = 0.000$)
Panel C: Group ATT (Heterogeneity by City Type)			
<i>Cluster 1: Dense Metropolis</i>			
Congestion Pricing (CP)	–	–0.3352* ($p = 0.061$)	–0.3966*** ($p = 0.007$)
Low Emission Zone (LEZ)	–	–0.1651* ($p = 0.089$)	–0.2299* ($p = 0.072$)
Synergy (CP $\times$ LEZ)	–	–1.4586*** ($p = 0.000$)	–1.3881*** ($p = 0.000$)
<i>Cluster 0: Sprawling Cities</i>			
Congestion Pricing (CP)	–	–0.2717*** ($p = 0.003$)	–0.3725*** ($p = 0.001$)
Low Emission Zone (LEZ)	–	–0.0508 ($p = 0.586$)	–0.0942 ($p = 0.402$)
Synergy (CP $\times$ LEZ)	–	–0.7792*** ($p = 0.002$)	–0.4570 ($p = 0.184$)

Notes: p -values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. 'Failed' estimation bounds denote overlap violations.

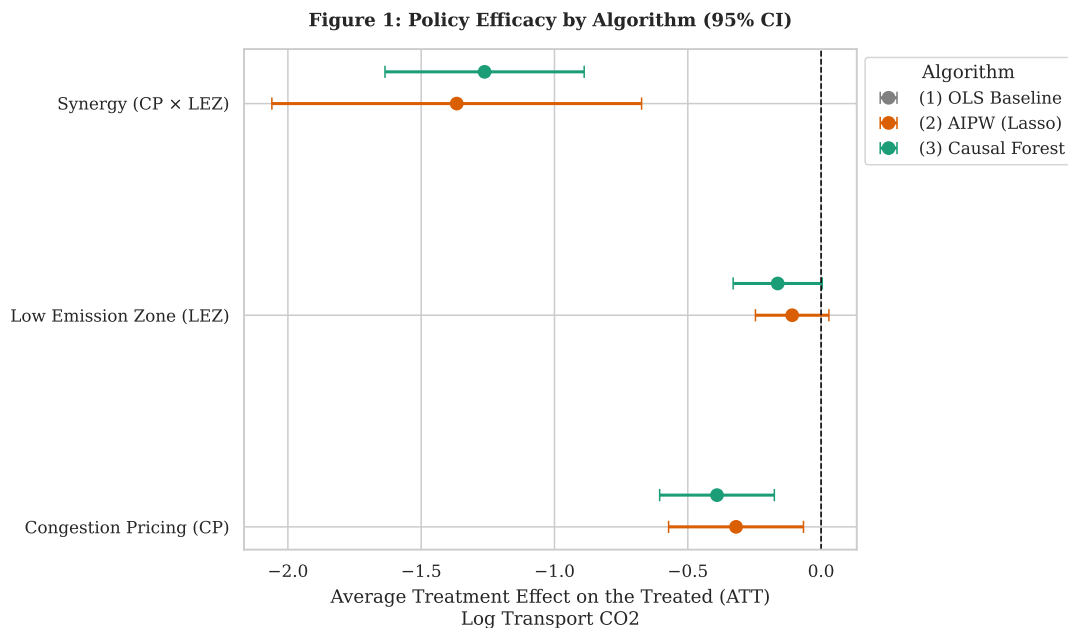


Figure 3: Policy Efficacy by Algorithm (95% CI)

7 Results

7.1 Spatial Heterogeneity

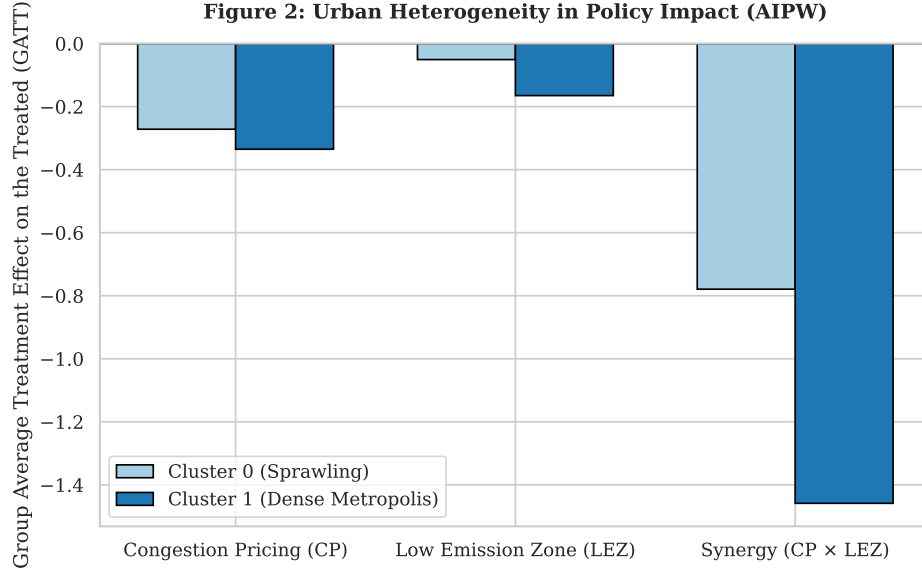


Figure 4: Urban Heterogeneity in Policy Impact (AIPW)

8 Sensitivity Analysis

Table 3: Propensity Score Distribution and Common Support

Statistic	Treated (CP Active)	Untreated (No CP)
Observations (N)	661	2450
Mean	0.569	0.105
Std. Dev.	0.264	0.161
Minimum	0.041	0.000
Median	0.603	0.031
Maximum	0.995	0.971

Notes: The common support region is strictly defined as $[0.0412, 0.9714]$. Within this bounded interval, the analysis retains 633 treated units (95.8%) and 1109 untreated units (45.3%).

Table 4: Falsification Tests: Estimated Effects on Placebo Outcomes (Global ATT)

Decoy Outcome	Congestion Pricing (CP)	Low Emission Zone (LEZ)	Synergy (CP $\times$ LEZ)
Panel A: AIPW (Lasso)			
No. of Libraries	−0.0424 ($p = 0.201$)	0.0601 ($p = 0.264$)	0.1164 ($p = 0.599$)
Streetlight Density	0.8594 ($p = 0.356$)	−0.5827 ($p = 0.537$)	−0.6432 ($p = 0.704$)
No. of Fountains	−0.2954 ($p = 0.414$)	−0.0395 ($p = 0.874$)	−0.2673 ($p = 0.519$)
No. of benches pc	0.0053 ($p = 0.964$)	−0.4626*** ($p = 0.001$)	−0.2084 ($p = 0.135$)
Panel B: Causal Forest			
No. of Libraries	−0.0701 ($p = 0.126$)	0.1036** ($p = 0.050$)	0.0092 ($p = 0.910$)
Streetlight Density	1.6208* ($p = 0.078$)	0.0816 ($p = 0.907$)	0.6502 ($p = 0.528$)
No. of Fountains	−0.4659 ($p = 0.173$)	−0.2444 ($p = 0.365$)	−0.3245 ($p = 0.462$)
No. of benches pc	−0.0436 ($p = 0.675$)	−0.5258*** ($p = 0.000$)	−0.3807** ($p = 0.010$)

Notes: p -values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

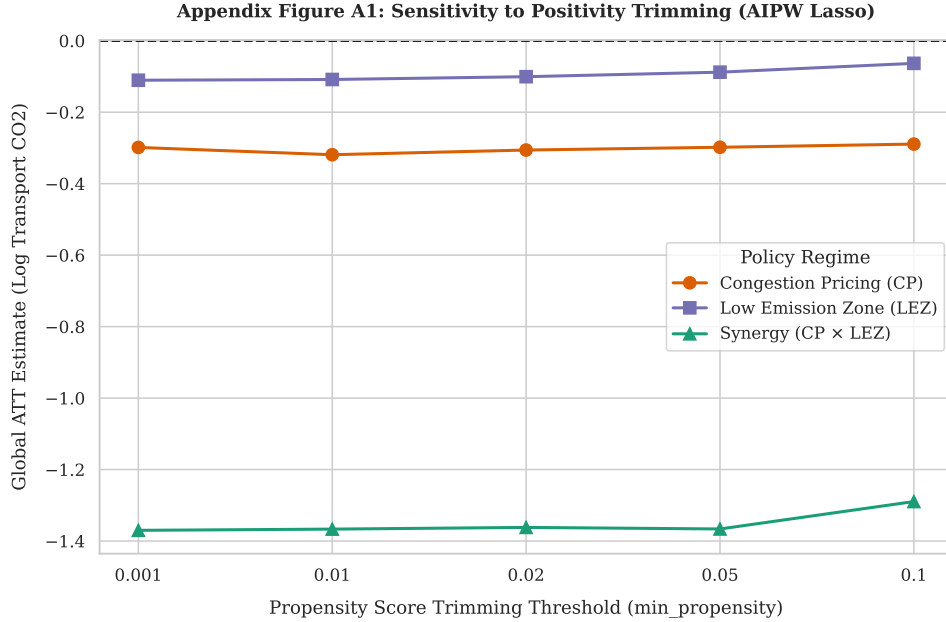


Figure 5: Sensitivity to Positivity Trimming

9 Interpretation and Discussion

Appendix

Declaration of Autorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.

Ferdinand Loser